KRISTY GOURAB SINHA

Department of Materials and Metallurgical Engineering
Bangladesh University of Engineering and Technology (BUET)

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EDUCATION

Bangladesh University of Engineering & Technology (BUET) Nov 2024 - Ongoing
M.Sc. in Materials Science Dhaka, Bangladesh
Department of Materials and Metallurgical Engineering
CGPA: 3.50/4.00

Military Institute of Science and Technology (MIST) 2024
B.Sc. in Nuclear Science and Engineering Dhaka, Bangladesh
Department of Nuclear Science and Engineering
CGPA: 3.67/4.00

Dhaka College 2019
Higher Secondary Certificate (H.S.C.) Dhaka, Bangladesh
Science Group
GPA: 5.00/5.00

WORK EXPERIENCE

- **Research Assistant (RA)**, Department of Nuclear Science and Engineering
  Military Institute of Science and Technology (MIST) Dec 2024 - Oct 2026
  - Research on projects funded by the MIST R&D Wing: **(i)** unmanned aerial vehicle for uranium ore exploration, nuclear radiation surveillance and atmospheric dispersion modeling **(ii)** cut-resistant fabric and STF-based bulletproof vests **(iii)** water-repellent surface functionalization of jute fabric for packaging application.
  - Assisting B.Sc. students with **projects** and **thesis**.
  - Supporting **SEM** and **WDXRF** analysis in the department laboratory.

RESEARCH PROJECTS & THESIS

- Research Project (2026, **CO-PI**): Development of Unmanned Aerial Vehicle for Uranium Ore Exploration, Nuclear Radiation Mapping and Atmospheric Dispersion Modeling
  Funded by: **MIST R&D Wing Research Grant, Centre for Army Research and Development** (2026)
- Research Project (2025, **CO-PI**): Cut-resistant Fabric and Shear Thickening Fluid (STF) based Bulletproof Vests
  Funded by: **MIST R&D Wing Research Grant, Centre for Army Research and Development** (2025)
- Pilot Study (2025): Gamma-ray Spectrometric Measurement of Natural Radionuclides (^{226}Ra, ^{232}Th, ^{40}K) in Tobacco Samples using an HPGe Detector
- Research Project (2024): Development of Advanced Cost-Effective Water-Repellent Surface Functionalization Strategies for Jute Fabric for Packaging Application
  Funded by: **MIST R&D Wing Research Grant, Centre for Army Research and Development** (2025)
- **B.Sc. Thesis** (Mar 2023 - Feb 2024): Synthesis and Characterization of Dysprosium Doped Lithium Strontium Borate for Dosimetry Application. **Supervisor: Dr. Md. Al-Mamun**, **Bangladesh Atomic Energy Commission**
  Funded by: **Ministry of Science and Technology, Bangladesh, Special Grant** (2023)
- **B.Sc. Project** (Jan 2022 - Jan 2023): Artificial Neural Network Based Nuclear Facility Inspection System and Computerized Exploration System for Radiological Ground Surveys. **Supervisor: Lt Col Faisal Kader**, Department of NSE, MIST

PUBLICATIONS

1. F. S. Siddique, **K. G. Sinha**, S. I. Safin, S. A. Islam, M. N. Hassan, M. A. Hasnat, S. Hossain, A. K. M. Rahman, M. Al-Mamun, "Exploring Dysprosium Doped Lithium Strontium Borate Based Phosphor for Dosimetry Application," Journal of Alloys and Compounds, 2025, 1041, 183628. DOI: 10.1016/j.jallcom.2025.183628
2. M. M. Rahman, **K. G. Sinha**, M. A. R. Alif, M. S. Hassan, et al., "Development of Water-Repellent Jute Fabric as a Sustainable Solution for Bag Production," BioResources, 2025, 21, 1001–1012. DOI: 10.15376/biores.21.1.1001-1012
3. M. U. Shafoyat, S. M. Al-Amin, B. Mogumdar, **K. G. Sinha**, M. E. Hoque, "Consumer Electronics and Gadgets Applications of Self-Healing Polymer Nanocomposites," Chapter 16 in M. E. Hoque, C. M. Hussain (Eds.), Smart Self-Healing Polymer Nanocomposites: Processes, Performance and Applications, Woodhead Publishing Series in Composites Science and Engineering, Woodhead Publishing (Elsevier), 2027, pp. 361–397. DOI: 10.1016/B978-0-443-40545-7.00013-8
4. A. A. M. Anik, M. S. A. Moon, A. T. Chowdhury, **K. G. Sinha**, S. I. Safin, F. S. Siddique, M. Z. H. Bhuiyan, T. Alam, "Computational Cognitive Network-Based Radiological Mechanism Inspection Technology and Digitally Accessible High-Tech Investigation Robot for Radiological Ground Surveys, Nuclear Environment Inspection, and Radioactive Substance Screening," in Advances in Machinery, Materials Science and Engineering Application XII, Advances in Transdisciplinary Engineering, IOS Press, 2025, pp. 477–486. DOI: 10.3233/ATDE251018
5. S. Mahmud, M. R. Iqbal, M. S. Ishrak, **K. G. Sinha**, "A Computational Study of NACA 23018 Airfoil Using Single and Double Rotating Cylinders as a Flow Control Device," AIP Conference Proceedings, 2026, 3451, 120006. DOI: 10.1063/5.0328306
6. S. Mahmud, M. R. Iqbal, M. S. Ishrak, **K. G. Sinha**, "Aerodynamic Performance of NACA 23018 Airfoil with Double Rotating Cylinders as Moving Surface Boundary-Layer Control (MSBC) Device at Multiple Reynolds Number: A Computational Study," AIP Conference Proceedings, 2026, 3451. DOI: 10.1063/5.0328309
7. M. M. R. Reyad, S. Islam, N. Zahin, M. T. R. Shaon, **K. G. Sinha**, M. A. Hossain, "An Enhanced YOLOv8 Framework for Real-Time Weapon Detection and Threat Assessment in Nuclear Power Plant Perimeter Security," in International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII), 2026.

CONFERENCE PRESENTATIONS

- PECCII 2026, Paper Presentation: "An Enhanced YOLOv8 Framework for Real-Time Weapon Detection and Threat Assessment in Nuclear Power Plant Perimeter Security," International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure, 2026
- ISMEE 2024, Paper Presentation: "A Convolutional Neural Network Approach for Steel Surface Defect Detection in Nuclear Facilities," 1st International Symposium on Materials, Energy, and Environment, Department of Chemistry, University of Chittagong, 2024
- ICEP 2022, Paper Presentation: "Artificial Neural Network Based Nuclear Facility Inspection System and Computerized Exploration System for Radiological Ground Surveys," 4th International Conference on Energy and Power, MIST, Dhaka, 2022

INDUSTRIAL TRAINING

- Non-Destructive Testing (NDT) Division, Atomic Energy Centre, Dhaka (AECD)
  Bangladesh Atomic Energy Commission, Shahbag, Dhaka 2023
- Centre for Research Reactor (CRR), Atomic Energy Research Establishment (AERE)
  Bangladesh Atomic Energy Commission, Savar, Dhaka 2023
- Rooppur Nuclear Power Plant
  Rooppur, Ishwardi, Pabna 2023

AWARDS & ACHIEVEMENTS

- Winner **(Top Ten Innovators)**, **Bangladesh Innovation Fair 2026**, organized by the Ministry of Science and Technology, Government of Bangladesh, Dhaka, September 2026
- **Dean's List of Honor**, **Military Institute of Science and Technology (MIST)**, Dhaka, Bangladesh, 2021
- **Finalist**, **TEKNOFEST 2026**, organized by the T3 Foundation (Turkish Technology Team Foundation) and the Ministry of Industry and Technology, **Turkey**, 2026
- **Winner**, Bangladesh ICT and Innovation Fair, Dhaka, 2026
- **Runner-up**, "Nuclear Dilemma 2024" Competition, organized by the Information Center on Nuclear Energy, Dhaka, May 2024
- 1st **Runner-up** in ICEP 2022 Poster Presentation: "Computerized High-Tech Exploration Robot for Radiological Ground Surveys, Nuclear Area Observation, Radioactive Material Detection and Handling," 4th International Conference on Energy and Power, MIST, Dhaka, December 2022
- **Winner**, "Nuclear Sprint" Competition, organized by the Information Center on Nuclear Energy, Dhaka, March 2021
- **3rd Place**, Project Display Event (Mechanical), 6th National Science Exposition 2018, Dhaka College Science Club, Dhaka College
- **Award**, Eco-friendly Project Event, ACC National Carnival 2018, Adamjee Cantonment Public School, Dhaka
- **Award**, Physical Project Event, **Notre Dame Annual Science Festival 2016 and 26th GKC**, Notre Dame Science Club, **Notre Dame College**, Dhaka
- **3rd Place**, Chemistry Project Event, 9th ISTARC Science and Technology Festival 2016, Ideal Science and Technology Aiming Research Council (ISTARC)

PROFESSIONAL MEMBERSHIP & LEADERSHIP

- Associate Member (A-29710), **Institution of Engineers**, Bangladesh (IEB)
- President, MIST Nuclear Engineering Club, Department of NSE, MIST — Mar 2023 – Mar 2024
- Vice President, Dhaka College Science Club — Jan 2018 – Jan 2019

KNOWLEDGE & SKILLS

1. Materials Characterization: XRD, SEM and WDXRF analysis
2. Simulation and Modeling: Materials Studio (DFT calculations using CASTEP), Geant4, PCTRAN, WinREMS
3. Data Processing and Image Analysis: Origin, ImageJ
4. Programming Languages: Python, C
5. Development Environment: Anaconda, Google Colab, Arduino IDE
6. Documentation and Design Software: Microsoft Office, Adobe Illustrator

LANGUAGE PROFICIENCY

- English: Excellent (speaking, reading and writing)

REFERENCE

Dr. Md. Al-Mamun
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University of Dhaka, Dhaka, Bangladesh
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